

LAB ASSIGNMENT – 1

DATE: 05/01/2026

Q1. Create a database named “dbMOURIMA_326” and switch to it.

Query: use dbMOURIMA_326

Output:

switched to db dbMOURIMA_326

Q2. Create a collection named students in the “dbMOURIMA_326” database.

Query: db.createCollection("students")

Output: { ok: 1 }

Q3. Insert one document into the students collection with the following fields:

- name
- rollNo
- department
- age

Query: db.students.insertOne({

... name: "Sujoy",

... rollNo: 1,

... department: "CSE",

... age: 20

... })

Output: {

acknowledged: true,

insertedId: ObjectId('69635072d44c36fc1a1e2621')

}

Q4. Insert four more documents into the students collection with different values.

Query: db.students.insertMany([

... { name: "Raj", rollNo: 2, department: "ECE", age: 21 },

... { name: "Sourav", rollNo: 3, department: "CSE", age: 19 },

... { name: "Sukanya", rollNo: 4, department: "ME", age: 22 },

... { name: "Ananya", rollNo: 5, department: "CSE", age: 20 }])

```
Output: {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('696350ddd44c36fc1a1e2622'),
    '1': ObjectId('696350ddd44c36fc1a1e2623'),
    '2': ObjectId('696350ddd44c36fc1a1e2624'),
    '3': ObjectId('696350ddd44c36fc1a1e2625')
  }
}
```

Q5. Display all documents present in the students collection.

Query: db.students.find()

```
Output: [{
  _id: ObjectId('69635072d44c36fc1a1e2621'),
  name: 'Sujoy',
  rollNo: 1,
  department: 'CSE',
  age: 20
},
{
  _id: ObjectId('696350ddd44c36fc1a1e2622'),
  name: 'Raj',
  rollNo: 2,
  department: 'ECE',
  age: 21
},
{
  _id: ObjectId('696350ddd44c36fc1a1e2623'),
  name: 'Sourav',
  rollNo: 3,
  department: 'CSE',
  age: 19},
}
```

```
{
  _id: ObjectId('696350ddd44c36fc1a1e2624'),
  name: 'Sukanya',
  rollNo: 4,
  department: 'ME',
  age: 22
},
{
  _id: ObjectId('696350ddd44c36fc1a1e2625'),
  name: 'Ananya',
  rollNo: 5,
  department: 'CSE',
  age: 20
}]
```

Q6. Display only those students who belong to the CSE department.

Query: `db.students.find({department: "CSE" })`

Output:

```
[{
  _id: ObjectId('69635072d44c36fc1a1e2621'),
  name: 'Sujoy',
  rollNo: 1,
  department: 'CSE',
  age: 20
},
{
  _id: ObjectId('696350ddd44c36fc1a1e2623'),
  name: 'Sourav',
  rollNo: 3,
  department: 'CSE',
  age: 19
},
```

```
{
  _id: ObjectId('696350ddd44c36fc1a1e2625'),
  name: 'Ananya',
  rollNo: 5,
  department: 'CSE',
  age: 20
}
```

Q7. Update the age of one student (any one) to a new value.

```
Query: db.students.updateOne(
... { name: "Sukanya" },
... { $set: { age: 23 } }
... )
db.students.find({ name: "Sukanya" }).pretty()
```

```
Output: {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

```
[ {
  _id: ObjectId('696350ddd44c36fc1a1e2624'),
  name: 'Sukanya',
  rollNo: 4,
  department: 'ME',
  age: 23
}]
```

Q8. Delete one document from the students collection using any condition.

Query: `db.students.deleteOne({ rollNo: 2 })`

Output: `{ acknowledged: true, deletedCount: 1 }`

Query: `db.students.find().pretty()`

Output: `[{`

`_id: ObjectId('69635072d44c36fc1a1e2621'),`

`name: 'Sujoy',`

`rollNo: 1,`

`department: 'CSE',`

`age: 20`

`},`

`{`

`_id: ObjectId('696350ddd44c36fc1a1e2623'),`

`name: 'Sourav',`

`rollNo: 3,`

`department: 'CSE',`

`age: 19`

`},`

`{`

`_id: ObjectId('696350ddd44c36fc1a1e2624'),`

`name: 'Sukanya',`

`rollNo: 4,`

`department: 'ME',`

`age: 23 },`

`{`

`_id: ObjectId('696350ddd44c36fc1a1e2625'),`

`name: 'Ananya',`

`rollNo: 5,`

`department: 'CSE',`

`age: 20}]`